THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 13: **Our Environment** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is a biodegradable substance? 1

(a) vegetable peels **(b)** DDT **(c)** glass **(d)** plastic bags

2. In a food chain, the organisms that make food from sunlight are: 1

(a) herbivores **(b)** producers **(c)** decomposers **(d)** carnivores

3. According to the 10% law, the energy available to the next trophic level is: 1

(a) 90% **(b)** 1%

(c) 10% of the previous level **(d)** 50%

4. A student states: “Ozone at higher levels of the atmosphere shields the Earth from UV radiation.” The correct response is: 1

(a) Correct **(b)** Correct only at ground level

(c) Incorrect — it causes UV radiation **(d)** Cannot be determined

5. For the chemicals that deplete the ozone layer, Student A says CFCs; Student B says carbon dioxide. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The increase in the concentration of harmful chemicals at successive trophic levels is called: 1

(a) eutrophication **(b)** pollution

(c) decomposition **(d)** biological magnification

7. Which of these is an artificial ecosystem? 1

(a) crop field **(b)** lake **(c)** forest **(d)** pond

8. The organisms which break down dead remains are: 1

(a) producers **(b)** herbivores **(c)** consumers **(d)** decomposers

9. In the food chain grass → deer → lion, the lion is a: 1

(a) decomposer **(b)** producer

(c) herbivore **(d)** tertiary/secondary consumer (carnivore)

10. Ozone is formed from oxygen by the action of: 1

(a) UV radiation **(b)** radio waves

(c) visible light **(d)** infrared radiation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is an ecosystem? 2

12. Why are decomposers important in an ecosystem? 2

13. Give two examples of non-biodegradable substances. 2

14. A student says that the number of trophic levels in a food chain is usually unlimited. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is the 10% law? Explain with an example of a food chain. 3

16. How is ozone formed? Why is its depletion a cause of concern? Name the chemical responsible. 3

17. What is biological magnification? Why is the concentration of DDT highest in humans? 3

OR

Suggest three ways to reduce the problem of waste disposal. 3

18. Why are food chains generally short? Explain. 3

OR

Differentiate between biodegradable and non-biodegradable substances with examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is a food web? 5
(ii) Explain the flow of energy in an ecosystem.
(iii) Why is the flow of energy unidirectional?

OR

(i) What are trophic levels? 5
(ii) Draw (describe) a food chain with four trophic levels.
(iii) Name the components of an ecosystem.

20. **Case study:** In a pond ecosystem, the concentration of a pesticide was: water 0.02 ppm, phytoplankton 5 ppm, small fish 40 ppm, large fish 300 ppm, fish-eating birds 2000 ppm. 5
(i) Name the phenomenon.
(ii) Which organism is most affected?
(iii) Why?
(iv) Suggest one measure to prevent it.

OR

(i) What is the ozone layer? 5
(ii) How is it being depleted?
(iii) What steps were taken internationally?
(iv) Effects of UV on humans?

END OF QUESTION PAPER — Total: 40 Marks